



CONTACT

EMAIL
hsandmann@ieee.org

WEB
hsandmann.github.io

GITHUB
hsandmann

LINKEDIN
in/hsandmann

PROJECT
graphole.com

LOCATION
São Paulo, Brazil

RELOCATION
Open to international relocation — on-site

SKILLS

Languages

Java Python Go Kotlin C
C++ JavaScript SQL

Machine Learning & AI

Deep Learning Neural Networks
Computer Vision NLP PCA SVM
Clustering

Backend & Architecture

Spring Boot Spring Cloud
Microservices Design Patterns
Software Architecture

Data & Infrastructure

PostgreSQL Redis Kafka
RabbitMQ Docker Kubernetes
Spark MongoDB Neo4j

Domains

Banking & Fintech
Cryptography (PKI/HSM)
High-Performance Computing
POS/ERP

EDUCATION

D.Sc. • Electrical Engineering
Polytechnic School, USP • 2007–2012

M.Sc. • Electrical Engineering
Polytechnic School, USP • 2003–2006

B.Sc. • Computer Science
FEI • 1999–2002

LANGUAGES

Portuguese Native
English Advanced
German B1

Humberto Sandmann

Computer Scientist & Professor

PROFILE

Senior software architect and artificial intelligence engineer with 20+ years building high-stakes systems across banking, fintech, and retail — core banking, cryptography (PKI/HSM), high-performance computing, computer vision, and cloud-native microservices. D.Sc. in neural computing; also a university professor teaching software design, AI, and distributed systems.

EXPERIENCE

Founder • H-Sandmann

2006–2025

Independent consultancy in machine learning and software architecture.

Machine Learning Software Architecture

Co-Founder • SelsanTech

2012–2023

Co-founded a venture delivering machine-learning and software-architecture solutions.

Machine Learning Software Architecture

Senior Software Architect & Machine Learning Engineer • RockSpoon

2018–2019

Architected POS/ERP/payroll platforms and computer-vision/ML features for a restaurant-tech company.

POS ERP JavaEE PostgreSQL OpenCV Computer Vision Go Android Kotlin

Senior Consultant, Computational Optimization • Banco Votorantim

2015–2016

Return-on-equity optimization on high-performance computing infrastructure for a major bank.

RoE High-Performance Computing J2EE Spring Boot Jenkins Test Automation

Senior Consultant, Computational Optimization • TecBan

2011

Senior Software Engineer / Architect • ABN Amro Bank

2006–2007

Senior System Analyst • Politec

2005–2006

Senior Software Engineer / Architect • EverSystems

2004–2005

Senior System Analyst • Central Online

2002–2003

Senior System Analyst • Datamace

1999–2002

Computer Programmer • Heraeus Electro-Nite

1998–1999

ACADEMIC POSITIONS

Associate Professor • Insper

2018–present

Associate Professor • ESPM

2014–2025

Associate Professor • FIAP

2012–2015

Visiting Professor • Toledo Prudente Centro Universitário

2015

Associate Professor • Universidade Cruzeiro do Sul

2012–2013

Temporary Professor • Universidade Nove de Julho

2011

Teaching Assistant • Polytechnic School, University of São Paulo

2008–2011

Full academic CV, publications and teaching record at hsandmann.github.io